

A Compact-Invariant-Core Form of the Bounded-Orbit Hypothesis

A candidate full answer to Remark 3.6 of arXiv:2404.18261

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Status. Candidate full solution, likely valid. The proof is elementary and the orbit hull construction is classical in semigroup fixed-point theory; novelty of the explicit semihypergroup formulation requires expert review.

The question and the answer

Let K be a semitopological semihypergroup and let X be a Banach space. Theorem 3.5 of Bandyopadhyay [1] characterizes left amenability of $AP(K)$ by common fixed points of separately continuous, equicontinuous affine actions

$$\pi : K \times (X^*, w^*) \longrightarrow (X^*, w^*)$$

having a norm-bounded orbit. Remark 3.6 asks for an equivalent, more conventional replacement for that orbit condition.

There is an exact replacement: require a nonempty weak-* compact convex invariant subset of X^* . In fact this is equivalent to the original hypothesis before amenability or equicontinuity enters.

Theorem 1 (bounded-orbit/invariant-core equivalence). *Let π be a separately weak-* continuous affine action of a nonempty semitopological semihypergroup K on X^* . The following conditions are equivalent.*

- (i) *There is $u \in X^*$ for which $\mathcal{O}(u) = \{\pi(t, u) : t \in K\}$ is norm bounded.*
- (ii) *There is $u \in X^*$ for which $\mathcal{O}(u)$ is relatively weak-* compact.*
- (iii) *There is a nonempty weak-* compact convex set $C \subseteq X^*$ such that $\pi(s, C) \subseteq C$ for every $s \in K$.*

When (i) holds, one may take

$$C = \overline{\text{co}}^{w^*} \mathcal{O}(u).$$

Proof. The equivalence of (i) and (ii) is a useful special feature of dual spaces. If $\mathcal{O}(u)$ is norm bounded, it lies in a scalar multiple of the weak-* compact dual unit ball, so Banach–Alaoglu makes it relatively weak-* compact. Conversely, the weak-* closure of a relatively weak-* compact orbit is weak-* compact. Every evaluation $v \mapsto v(x)$ is therefore bounded on it for each $x \in X$, and the uniform boundedness principle gives a uniform norm bound.

Assume (i) and set $C = \overline{\text{co}}^{w^*} \mathcal{O}(u)$. The orbit is contained in some rB_{X^*} , so C is a weak-* closed subset of the weak-* compact set rB_{X^*} . Hence C is nonempty, weak-* compact, and convex.

It remains to prove invariance, which is the only semihypergroup-specific point. Fix $s, t \in K$. By the action axiom,

$$\pi(s, \pi(t, u)) = \int_K \pi(\zeta, u) d(p_s * p_t)(\zeta).$$

The convolution $p_s * p_t$ is a compactly supported probability measure. Since $\zeta \mapsto \pi(\zeta, u)$ is weak-* continuous, its integral is a weak-* barycenter of points of $\mathcal{O}(u)$ and consequently belongs to C . Thus $\pi(s, \mathcal{O}(u)) \subseteq C$. Affineness gives $\pi(s, \text{co } \mathcal{O}(u)) \subseteq C$, and weak-* continuity of π_s extends this inclusion to C . Therefore (iii) holds.

Finally assume (iii), choose $u \in C$, and use invariance to obtain $\mathcal{O}(u) \subseteq C$. A weak-* compact subset of X^* is norm bounded by the uniform boundedness principle, so (i) follows. $\square$

Corollary 2 (answer to the replacement question). *In Theorem 3.5 of [1], “admits an element with bounded orbit” may be replaced, without changing the class of representations, by either of the following:*

1. *the action admits a relatively weak-* compact orbit;*

2. the action admits a nonempty weak-* compact convex invariant subset.

In particular, the second condition is the usual compact invariant-domain formulation of an affine fixed-point property.

Proof. Apply Theorem 1 to each representation quantified over in Theorem 3.5. Equicontinuity and left amenability are not needed for the equivalence of the three orbit conditions; they remain exactly as in the source theorem. $\square$

The restriction cannot be omitted

The replacement above is exact, but no orbit condition can simply be erased from the theorem.

Proposition 3 (sharpness). *There is a semitopological semihypergroup K for which $AP(K)$ is left amenable and a jointly continuous, equicontinuous affine action of K on a dual Banach space having no common fixed point.*

Proof. Take the discrete semihypergroup associated with the additive group $K = \mathbb{Z}$, so $p_m * p_n = p_{m+n}$. The group $\mathbb{Z}$ is amenable: weak-* cluster points of its Følner averages give a translation-invariant mean on $\ell^\infty(\mathbb{Z})$, whose restriction is a left invariant mean on $AP(\mathbb{Z})$.

Identify $\mathbb{R}$ with the dual of the one-dimensional real Banach space and define

$$\pi(n, x) = x + n.$$

This is a jointly continuous affine action, and its coefficient maps are isometries, hence form an equicontinuous family. A common fixed point would satisfy $x + 1 = x$, which is impossible. Each orbit $x + \mathbb{Z}$ is unbounded, in agreement with Theorem 1. $\square$

Scope and provenance

The result fully answers the literal existence question in Remark 3.6: it gives two equivalent representation conditions, one of which is the standard compact convex invariant-domain hypothesis. It does not assert that this condition is computationally easier in every concrete example.

The use of a weak-* closed convex orbit hull is classical and occurs in the semigroup precursor of Lau and Zhang [2]. The new content claimed here is only the explicit three-way formulation and the barycentric verification for the semihypergroup action axiom. A bounded literature search found no explicit published statement presenting this as the answer to the source remark; priority should nevertheless be checked by a specialist.

References

- [1] C. Bandyopadhyay, *Common Fixed Points of Semihypergroup Representations*, arXiv:2404.18261, 2024.
- [2] A. T.-M. Lau and Y. Zhang, *Amenability properties and fixed point properties of affine representations of semigroups*, Proc. Amer. Math. Soc. 149 (2021), 4815–4823.